

...e's a **100-day DSA roadmap** tailored to your goals, combining the **DSA Master.pdf** (for foundational concepts) and **Interview Master 100.pdf** (for problem-solving), along with **FANG/MANG patterns**, **frameworks**, and **advanced topics**. The plan balances theory, practice, and revision.

Key Principles

1. Daily Structure

- **Theory (30-45 min)**: Learn 1-2 core concepts (from DSA Master).
- **Problem-Solving (1.5-2 hrs)**: Solve 2-3 LeetCode problems (from Interview Master 100 + FANG lists).
- **Patterns/Frameworks (30 min)**: Focus on 1 pattern/framework (e.g., Sliding Window, DP).
- **Revision (15 min)**: Revisit weak areas or past problems.

2. Weekly Focus

- **Days 1-20**: Foundations (Arrays, Strings, Linked Lists, Stacks/Queues).
- **Days 21-40**: Trees, Graphs, Heaps, Hashing.
- **Days 41-60**: DP, Greedy, Backtracking, Bit Manipulation.
- **Days 61-80**: Advanced Graphs, Tries, System Design Basics, FANG Problems.
- **Days 81-100**: Mock Interviews, Revision, and Weak Areas.

3. Resources

- **Primary**: DSA Master (theory), Interview Master 100 (problems).
- **Supplementary**:
 - [LeetCode Top 100](https://leetcode.com/explore/interview/card/top-interview-questions-medium/)
 - [Grokking the Coding Interview](https://www.educative.io/courses/grokking-the-coding-interview) (for patterns).
 - [NeetCode.io](https://neetcode.io/) (roadmap + solutions).
 - [Striver's SDE Sheet](https://takeuforward.org/interviews/strivers-sde-sheet-top-coding-interview-problems/) (FANG-focused).

100-Day Breakdown

*(Problems marked with **[IM100]** are from Interview Master 100; others are FANG/MANG additions.)*

Phase 1: Foundations (Days 1-20)

Goal: Master arrays, strings, linked lists, stacks/queues, and basic algorithms (Two Pointers, Sliding Window, Binary Search).

Day	Theory (DSA Master)	Problems (Interview Master 100 + FANG)
Pattern/Framework		
1	Arrays: Sorting (QuickSort/MergeSort)	[1. Two Sum](https://leetcode.com/problems/two-sum/) [IM100] , [Two Pointers]
2	Arrays: Binary Search	[8. Binary Search](https://leetcode.com/problems/binary-search/) [IM100] , [33. Search in Rotated Sorted Array](https://leetcode.com/problems/search-in-rotated-sorted-array/) Binary Search Tricks
3	Arrays: Two Pointers	[15. 3Sum](https://leetcode.com/problems/3sum/) [IM100] , [167. Two Sum II](https://leetcode.com/problems/two-sum-ii-input-array-is-sorted/) Two Pointers for Sorted Arrays
4	Arrays: Sliding Window	[3. Longest Substring Without Repeating Characters](https://leetcode.com/problems/longest-substring-without-repeating-characters/) [IM100] , [567. Permutation in String](https://leetcode.com/problems/permutation-in-string/) Sliding Window Template
5	Strings: Manipulation	[5. Valid Palindrome](https://leetcode.com/problems/valid-palindrome/) [IM100] , [125. Valid Palindrome II](https://leetcode.com/problems/valid-palindrome-ii/) Palindrome Checks
6	Linked Lists: Traversal/Insertion	[21. Merge Two Sorted Lists](https://leetcode.com/problems/merge-two-sorted-lists/) [IM100] , [206. Reverse Linked List]

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(https://leetcode.com/problems/reverse-linked-list/) | Linked List Reversal
| 7 | Linked Lists: Cycle Detection | [9. Linked List Cycle]
(https://leetcode.com/problems/linked-list-cycle/) | [IM100], [142. Linked List Cycle II]
(https://leetcode.com/problems/linked-list-cycle-ii/) | Floyd's Tortoise and Hare
| 8 | Stacks: LIFO Operations | [20. Valid Parentheses]
(https://leetcode.com/problems/valid-parentheses/) | [IM100], [155. Min Stack]
(https://leetcode.com/problems/min-stack/) | [IM100] | Stack for Parentheses/MonoStack
| 9 | Queues: FIFO Operations | [239. Sliding Window Maximum]
(https://leetcode.com/problems/sliding-window-maximum/) (Use Deque) | Monotonic Queue

| 10 | **Revision Day** | Re-solve 5 problems from Days 1-9
| Identify Weak Areas |
| 11 | Hash Tables: Collision Handling | [28. Contains Duplicate]
(https://leetcode.com/problems/contains-duplicate/) | [IM100], [387. First Unique Character]
(https://leetcode.com/problems/first-unique-character-in-a-string/) | Frequency Counting

| 12 | Hash Tables: Anagrams | [7. Valid Anagram]
(https://leetcode.com/problems/valid-anagram/) | [IM100], [49. Group Anagrams]
(https://leetcode.com/problems/group-anagrams/) | [IM100] | Anagram Grouping
| 13 | **Mock Test 1** | Solve 3 random problems from IM100 (Days 1-12)
| Time Yourself (45 mins) |
| 14 | Recursion: Basics | [24. Swap Nodes in Pairs]
(https://leetcode.com/problems/swap-nodes-in-pairs/) | [206. Reverse Linked List (Recursive)]
(https://leetcode.com/problems/reverse-linked-list/) | Recursive Linked List Operations
| 15 | Recursion: Backtracking | [46. Permutations]
(https://leetcode.com/problems/permutations/) | [IM100], [77. Combinations]
(https://leetcode.com/problems/combinations/) | Backtracking Template
| 16 | **Project Day** | Implement a **LRU Cache** (LeetCode 146)
| Combine HashMap + Doubly Linked List |
| 17 | Bit Manipulation: Basics | [83. Single Number]
(https://leetcode.com/problems/single-number/) | [IM100], [191. Number of 1 Bits]
(https://leetcode.com/problems/number-of-1-bits/) | XOR Tricks
| 18 | **Revision Day** | Re-solve 5 problems from Days 13-17
| Focus on Recursion/Bitwise |
| 19 | **Mock Test 2** | Solve 3 problems from LeetCode's "Top Interview
Questions" (Medium) | Time Yourself (1 hour)
| 20 | **Review & Plan Phase 2** | Analyze progress, adjust pace

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Phase 2: Trees, Graphs, Heaps (Days 21-40)

Goal: Master tree/graph traversals, heaps, and advanced patterns (DFS/BFS, Topological Sort).

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| **Day** | **Theory** | **Problems**
| **Pattern** |
|-----|-----|-----|
| 21 | Trees: Traversals (In/Pre/Postorder) | [10. Max Depth of Binary Tree]
(https://leetcode.com/problems/maximum-depth-of-binary-tree/) | [IM100], [104. Binary Tree Level
Order Traversal](https://leetcode.com/problems/binary-tree-level-order-traversal/) | [IM100] |
BFS/DFS Templates
| 22 | Trees: BST Validation | [26. Validate BST]
(https://leetcode.com/problems/validate-binary-search-tree/) | [IM100], [98. Validate BST (Follow-
up)](https://leetcode.com/problems/validate-binary-search-tree/) | BST Properties
| 23 | Trees: Construction | [35. Construct Binary Tree from Preorder and
Inorder](https://leetcode.com/problems/construct-binary-tree-from-preorder-and-inorder-traversal/)
| [IM100] | Divide and Conquer
| 24 | Heaps: Priority Queues | [40. Kth Largest Element in an Array]
(https://leetcode.com/problems/kth-largest-element-in-an-array/) | [IM100], [215. Kth Largest
Element](https://leetcode.com/problems/kth-largest-element-in-an-array/) | QuickSelect vs Heap
| 25 | Graphs: BFS/DFS | [76. Number of Islands]
(https://leetcode.com/problems/number-of-islands/) | [IM100], [200. Number of Islands]

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list cycle III]

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https://leetcode.com/problems/number-of-islands/) | Grid Traversal (DFS/BFS) |
26 | Graphs: Cycle Detection | [65. Course Schedule]
https://leetcode.com/problems/course-schedule/) `IM100` (Topological Sort) | Kahn's Algorithm
27 | Graphs: Shortest Path | [743. Network Delay Time]
https://leetcode.com/problems/network-delay-time/) (Dijkstra) | Priority Queue for Dijkstra
28 | **Revision Day** | Re-solve 5 problems from Days 21-27
Focus on Trees/Graphs
29 | **Mock Test 3** | Solve 3 problems from LeetCode's "Graph" tag
Time Yourself (1 hour)
30 | Tries: Implementation | [39. Implement Trie]
https://leetcode.com/problems/implement-trie-prefix-tree/) `IM100`, [208. Implement Trie]
https://leetcode.com/problems/implement-trie-prefix-tree/) | Trie for Autocomplete
31 | Union-Find | [92. Surrounded Regions]
https://leetcode.com/problems/surrounded-regions/) `IM100` | Union-Find for Grid Problems
32 | **Project Day** | Implement a **TinyURL System** (LeetCode 535)
Hashing + Encoding
33 | Heaps: Top K Elements | [53. Top K Frequent Elements]
https://leetcode.com/problems/top-k-frequent-elements/) `IM100`, [347. Top K Frequent Words]
https://leetcode.com/problems/top-k-frequent-words/) | Heap + HashMap
34 | **Revision Day** | Re-solve 5 problems from Days 29-33
Focus on Heaps/Tries
35 | **Mock Test 4** | Solve 3 problems from FANG interview lists (e.g.,
Google) | Time Yourself (1.5 hours)
36-40 | **Buffer Days** | Catch up on backlog or explore advanced graph
problems (e.g., A*, Bellman-Ford) |
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Phase 3: DP, Greedy, Backtracking (Days 41-60)

Goal: Master dynamic programming, greedy algorithms, and backtracking patterns.

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| **Day** | **Theory** | **Problems**
| **Pattern** |
|-----|-----|-----|
41 | DP: Fibonacci Sequence | [20. Climbing Stairs]
https://leetcode.com/problems/climbing-stairs/) `IM100`, [70. Climbing Stairs]
https://leetcode.com/problems/climbing-stairs/) | Memoization vs Tabulation
42 | DP: Knapsack Problem | [322. Coin Change]
https://leetcode.com/problems/coin-change/) `IM100`, [416. Partition Equal Subset Sum]
https://leetcode.com/problems/partition-equal-subset-sum/) | 0/1 Knapsack Template
43 | DP: Longest Subsequence | [72. Longest Increasing Subsequence]
https://leetcode.com/problems/longest-increasing-subsequence/) `IM100`, [300. LIS]
https://leetcode.com/problems/longest-increasing-subsequence/) | LIS with Binary Search Optimization
44 | DP: String Problems | [41. Longest Palindromic Substring]
https://leetcode.com/problems/longest-palindromic-substring/) `IM100`, [516. Longest Palindromic
Subsequence](https://leetcode.com/problems/longest-palindromic-subsequence/) | DP on Strings
45 | Greedy: Interval Scheduling | [30. Merge Intervals]
https://leetcode.com/problems/merge-intervals/) `IM100`, [56. Merge Intervals]
https://leetcode.com/problems/merge-intervals/) | Interval Merging
46 | Greedy: Huffman Coding | [621. Task Scheduler]
https://leetcode.com/problems/task-scheduler/) | Priority Queue for Greedy
47 | Backtracking: Subsets | [52. Subsets]
https://leetcode.com/problems/subsets/) `IM100`, [78. Subsets]
https://leetcode.com/problems/subsets/) | Subset Generation Template
48 | Backtracking: Permutations | [46. Permutations]
https://leetcode.com/problems/permutations/) `IM100` (Revisit) | Permutation Template
49 | **Revision Day** | Re-solve 5 DP/Greedy problems
Focus on State Transition |
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| 50 | **Mock Test 5** | Solve 3 DP problems from LeetCode's "Dynamic
Programming" tag | Time Yourself (1.5 hours) |
| 51-60 | **Advanced DP/Backtracking** | Solve 1-2 problems daily from:
- [LeetCode DP Tag](https://leetcode.com/tag/dynamic-programming/)
- [Backtracking Tag](https://leetcode.com/tag/backtracking/)
- FANG Lists (e.g., [Facebook DP Problems](https://leetcode.com/discuss/interview-
experience/1246512/Facebook-Dynamic-Programming-Problems)) | Create a DP Cheat Sheet

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Phase 4: Advanced Topics & FANG Problems (Days 61-80)

****Goal****: Tackle advanced graphs, system design basics, and FANG/MANG interview questions.

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| **Day** | **Topic** | **Problems**
| **Focus** |
|-----|-----|-----|
| 61 | Advanced Graphs: Dijkstra's | [787. Cheapest Flights Within K Stops]
| (https://leetcode.com/problems/cheapest-flights-within-k-stops/) | Dijkstra with Limited Steps
| 62 | Advanced Graphs: Bellman-Ford | [743. Network Delay Time]
| (https://leetcode.com/problems/network-delay-time/) (Revisit) | Negative Weight Handling
| 63 | Advanced Graphs: A* Algorithm | [2093. Minimum Cost to Reach City With
Discounts](https://leetcode.com/problems/minimum-cost-to-reach-city-with-discounts/) | Heuristic
Search
| 64 | System Design Basics | Study: [System Design Primer]
| (https://github.com/donnemartin/system-design-primer) | Scalability, Caching, Load Balancing
| 65 | Bit Manipulation: Advanced | [201. Bitwise AND of Numbers Range]
| (https://leetcode.com/problems/bitwise-and-of-numbers-range/) `[IM100]`, [338. Counting Bits]
| (https://leetcode.com/problems/counting-bits/) | Bitmasking Tricks
| 66 | Math: Probability/Combinatorics | [62. Unique Paths]
| (https://leetcode.com/problems/unique-paths/) `[IM100]`, [63. Unique Paths II]
| (https://leetcode.com/problems/unique-paths-ii/) | Combinatorics in DP
| 67 | **Mock Interview 1** | Use [Pramp](https://www.pramp.com/) or
[Interviewing.io](https://interviewing.io/) | Simulate Real Interview
| 68-70 | **FANG Problem Sets** | Solve 2 problems daily from:
- [Google](https://leetcode.com/discuss/interview-experience/1246512/Google-Top-Interview-Problems)
- [Amazon](https://leetcode.com/discuss/interview-experience/1246512/Amazon-Top-Interview-Problems)
- [Facebook/Meta](https://leetcode.com/discuss/interview-experience/1246512/Facebook-Top-Interview-
Problems) | Pattern Recognition
| 71 | **Revision Day** | Re-solve 5 FANG problems
| Time Optimization
| 72 | **Mock Interview 2**

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